

Thirty-thousand-year-old distant relative of giant icosahedral DNA viruses with a pandoravirus morphology

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The largest known DNA viruses infect *Acanthamoeba* and belong to two markedly different families. The Megaviridae exhibit pseudo-icosahedral virions up to 0.7 μm in diameter and adenine-thymine (AT)-rich genomes of up to 1.25 Mb encoding a thousand proteins. Like their Mimivirus prototype discovered 10 y ago, they entirely replicate within cytoplasmic virion factories. In contrast, the recently discovered Pandoraviruses exhibit larger amphora-shaped virions 1 μm in length and guanine-cytosine-rich genomes up to 2.8 Mb long encoding up to 2,500 proteins. Their replication involves the host nucleus. Whereas the Megaviridae share some general features with the previously described icosahedral large DNA viruses, the Pandoraviruses appear unrelated to them. Here we report the discovery of a third type of giant virus combining an even larger pandoravirus-like particle 1.5 μm in length with a surprisingly smaller 600 kb AT-rich genome, a gene content more similar to Iridoviruses and Marseillevirus, and a fully cytoplasmic replication reminiscent of the Megaviridae. This suggests that pandoravirus-like particles may be associated with a variety of virus families more diverse than previously envisioned. This giant virus, named *Pithovirus sibericum*, was isolated from a >30,000-y-old radiocarbon-dated sample when we initiated a survey of the virome of Siberian permafrost. The revival of such an ancestral amoeba-infecting virus used as a safe indicator of the possible presence of pathogenic DNA viruses, suggests that the thawing of permafrost either from global warming or industrial exploitation of circumpolar regions might not be exempt from future threats to human or animal health.

giant DNA virus | late Pleistocene | icosahedral capsid

Ten years ago, the discovery of *Acanthamoeba polyphaga* Mimivirus revealed the existence of giant DNA viruses with particles large enough to be visible under a light microscope (1, 2). Further sampling of various environments and geographical locations led to the isolation of Mimivirus variants (3, 4) and more distant relatives, two of which have been fully sequenced: Moumouvirus (5) and Megavirus chilensis (6). All of these viruses share the same distinctive particle structure: a unique external fiber layer enclosing a pseudoicosahedral protein capsid of about 0.5 μm in diameter, itself containing lipid membranes surrounding an electron-dense nucleocapsid. They share an adenine-thymine (AT)-rich (>70%) linear DNA genome with sizes up to 1.26 Mb encoding up to 1,120 proteins (6). They all encode a full transcription apparatus allowing them to replicate in the host's cytoplasm. These common features suggested that all giant viruses were to belong to a single family (Megaviridae) sharing the same particle morphology, genomic features, and replication strategy (7, 8).

This assumption was radically challenged by the discovery of Pandoraviruses (9), which show different and somewhat opposite characteristics. These Acanthamoeba-infecting viruses exhibit much

larger amphora-shaped virions 1–1.2 μm in length. Their guanine-cytosine (GC)-rich (>61%) genomes are up to 2.8 Mb long and encode up to 2,500 proteins sharing no resemblance with those of Megaviridae (9). Finally, Pandoravirus particles do not incorporate the transcription machinery that would allow them to entirely replicate in the host's cytoplasm. Known giant viruses infecting *Acanthamoeba* were thus thought to belong to two very dissimilar types in terms of particle structure, genome characteristics, and replication strategies. Here we describe a third type of giant virus named “Pithovirus” (from the Greek word *pitios* designating the kind of large amphora handed over by the gods to the legendary Pandora) propagating in an even larger pandoravirus-like particle, but exhibiting a replication cycle and genomic features reminiscent of those of large icosahedral nucleocytoplasmic DNA viruses. Giant viruses are thus much more diverse than initially assumed.

Significance

Giant DNA viruses are visible under a light microscope and their genomes encode more proteins than some bacteria or intracellular parasitic eukaryotes. There are two very distinct types and infect unicellular protists such as *Acanthamoeba*. On one hand, Megaviridae possess large pseudoicosahedral capsids enclosing a megabase-sized adenine-thymine-rich genome, and on the other, the recently discovered Pandoraviruses exhibit micron-sized amphora-shaped particles and guanine-cytosine-rich genomes of up to 2.8 Mb. While initiating a survey of the Siberian permafrost, we isolated a third type of giant virus combining the Pandoravirus morphology with a gene content more similar to that of icosahedral DNA viruses. This suggests that pandoravirus-like particles may correspond to an unexplored diversity of unconventional DNA virus families.

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Data deposition: The Pithovirus genome sequence reported in this paper has been deposited in GenBank database (accession no. KF740664). The MS proteomics data reported in this paper have been deposited in the ProteomeXchange Consortium database, <http://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PX0000460>, via the Proteomics Identifications database (www.ebi.ac.uk/pride/archive) partner repository [identifier no. PX0000460 (10.6019/PX0000460)].

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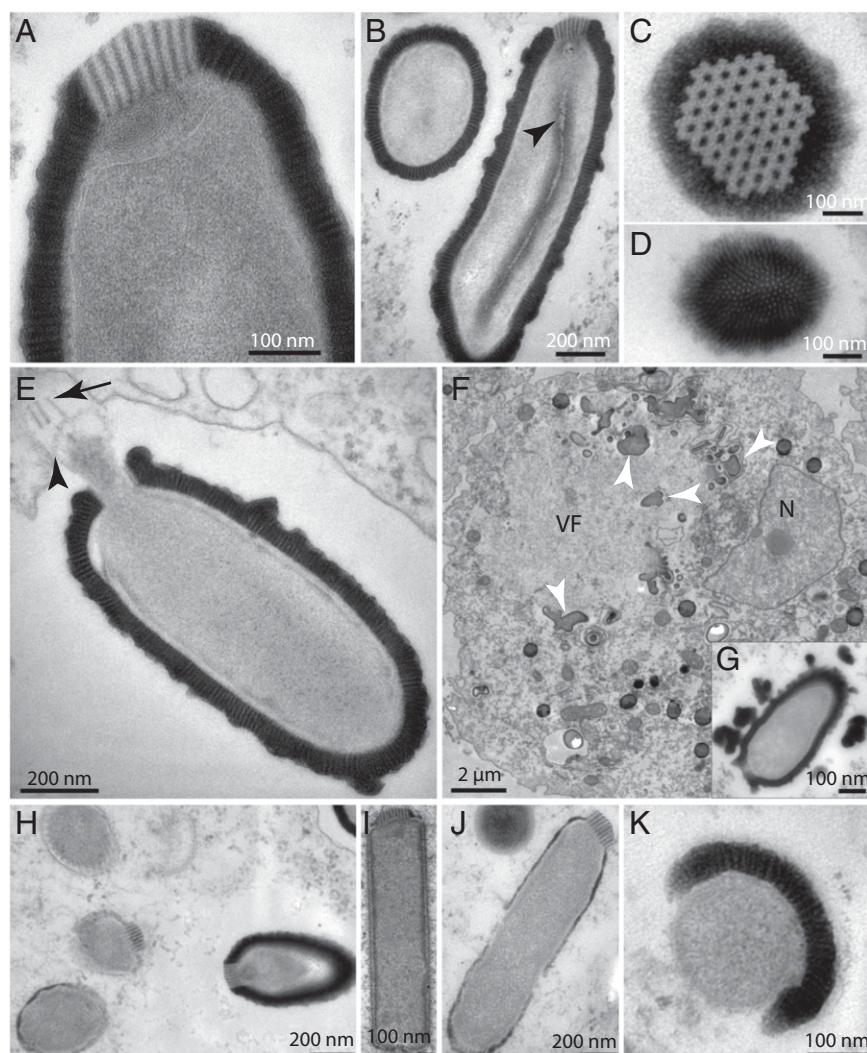


Fig. 1. Electron microscopy imaging of the Pithovirus replication cycle in *A. castellanii*. (A) Apex of the Pithovirus particle showing its unique cork made of 15 nm-spaced stripes, rolled membranes underneath, and the internal membrane. (B) Two perpendicular views of the Pithovirus particles (cross and longitudinal sections). The particles are wrapped into a 60 nm-thick envelope made of 10 nm-spaced parallel stripes. A lipid membrane is enclosing a homogeneous interior where a tubular structure is seen episodically, but in a reproducible fashion (arrowhead). (C) Top view of the cork revealing a hexagonal honeycomb-like array. (D) Bottom view of the particle showing the striated organization of the envelope. (E) An opened Pithovirus particle in the host vacuole. Parts of the expelled cork are visible (black arrows) and the internal membrane of the particle (black arrowhead) appears ready to fuse with the vacuole membrane. (F) Maturing virions at a late stage of infection. Structures made of stripes, pieces of cork, and dense material accumulate (white arrowhead) in the periphery of the virion factory (VF). These structures may contain preassembled particle building blocks (Fig. S1). The cell nucleus (N) is visible. (G) Inset highlighting a late stage of virion maturation with globular striated structures accumulating at the virion periphery. (H) Various stages of particle assembly in the same cell. (I) Incompletely assembled rectangular particle lacking its thick envelope. The striated cork is already visible. (J) At a later stage, the particle adopts its final rounded shape while its envelope thickens. (K) Orthogonal view of an immature virion showing the envelope in the process of wrapping the particle.

Results

Particle Morphology. If the notion that DNA may persist over geological timespans (>1 My) is gaining a progressive acceptance, the fact that cellular organisms might survive that long remains a contentious issue (10, 11). Because of its neutral pH and reducing and anaerobic properties, northeast Siberian permafrost is among the most suitable environments to look for long-term surviving microorganisms (12) or even plants (13). Except for few studies targeting the Influenza virus and the Variola virus over historical timespans (14, 15), the possibility that DNA viruses might remain infectious over a much longer time scale has not yet been investigated. We recently initiated such a survey using *Acanthamoeba* as bait and its giant DNA viruses as safe surrogates for pathogenic viruses. In this context, Pithovirus was initially spotted using light microscopy as ovoid particles (Fig. S1) multiplying in a culture of *Acanthamoeba castellanii* inoculated with a sample of Siberian permafrost from the Kolyma lowland region (SI Materials and Methods). This sample was aseptically collected from a permafrost layer corresponding to late Pleistocene sediments older than 30,000 y (13). Similar paleosoils are known to contain cysts of *Acanthamoeba* cells (16). After amplification, the particles were analyzed by transmission electron microscopy confirming that Pithovirus shared the overall morphology of the Pandoraviruses (9) with slightly larger dimensions (~1.5 μ m in length, 500 nm in diameter). The virions exhibit a 60 nm-thick structured envelope made of one layer of parallel stripes (Fig. 1). An internal membrane (Fig. 1A) encases a compartment without discernible substructures

except for an electron-dense sphere (50 nm in diameter) seen episodically but in a reproducible fashion (Fig. S1) and a tubular structure parallel to the long axis of the particle (Fig. 1B). At variance with the Pandoraviruses, the apex aperture of the Pithovirus particle appears sealed by a protruding cork (80 nm thick and 160 nm wide) with a hexagonal grid structure (Fig. 1A and C) reminiscent of the organization of capsomers in icosahedral virions. A coil of rolled-up membrane seems to be connected to this apex structure (Fig. 1A).

Replication Cycle. The Pithovirus replication strategy was documented by following its propagation in axenic *Acanthamoeba* cultures over an entire multiplication cycle, starting from purified particles. A complete lysis of infected cultures occurred in 10–20 h depending on the initial number of virus particles. As for Pandoraviruses, the replication cycle begins with the phagocytosis of individual particles. First, the Pithovirus particles lose their apical cork allowing the underlying lipid membrane to fuse with the cellular vacuole membrane (Fig. 1E). This creates a channel between the most internal compartment of the virion and the cell cytoplasm (9). In contrast with the Pandoraviruses, the cell nucleus maintains its shape throughout the entire Pithovirus replication cycle (Movie S1). The first visible sign of infection is the formation of an area cleared of cytoplasm subcellular structures, 4–6 h postinfection (Fig. 1F). Numerous vesicles start accumulating in this presumable virion factory (Fig. S1). The process of virion formation is reminiscent of

Pandoravirus, the envelope and the interior of the Pithovirus particles being assembled or “knitted” simultaneously (9) (Fig. 1*H* and [Movie S1](#)). First, rectangular-shaped closed particles with their characteristic cork appear at the periphery of the virion factory (Fig. 1*I*). Later on, their outer wall thickens and the particles take their final ovoid shape, still lacking their thick striated tegument (Fig. 1*J*). This layer is built subsequently and piecewise, as evidenced in cross section images of immature particles (Fig. 1*K*) as well as by the fuzzy appearance of the tegument-like envelope in the latest stages of particle maturation (Fig. 1*G* and [Fig. S1](#)). After 6–8 h, particles at various stages of maturation may coexist in the same virion factory.

Besides dense vesicles of unknown composition accumulating inside the infected cells, shapeless blobs made of pieces of striated envelope, pieces of the corks, and diffuse material reminiscent of the mature particles interior (Fig. 1*F* and [Fig. S1](#)) are seen at the periphery of the virion factories. They might be reservoirs of partially organized virion building blocks. Mature particles are found in equal amounts in the cytoplasm or in vacuoles suggesting that they could exit the cell by exocytosis ([Movie S1](#)). The replicative cycle ends with the cells releasing hundreds of particles upon lysis. Despite their bacterial-like dimensions no image of particles undergoing binary fission was encountered throughout our comprehensive electron microscopy study, hinting at the viral nature of Pithoviruses before the analysis of their gene content.

Genome Sequencing. DNA from purified particles was sequenced using the standard Illumina protocol (2× 100 nt paired end reads). The resulting 600 kb of unique sequence data readily allowed the identification of most genes, but could not be assembled in less than 80 nonoverlapping scaffolds. The finished genome was obtained by sequencing a Nextera mate pair library (5–8 kb inserts) using the Illumina MiSeq platform (1,066,320 reads, 2× 250 nt) combined with 77,241 PacBio RS long single-end reads. Unexpectedly, given the similar morphology of their particles, the Pithovirus genome was found to be completely different from the Pandoravirus’s in terms of size, nucleotide composition, topology, and gene content. The Pithovirus genome consists of an AT-rich (64%) dsDNA molecule of a mere 610,033 bp whereas the Pandoraviruses exhibit a GC-rich genome (> 61%) of up to 2.8 Mb. As for the Iridoviridae (17) and the Acanthamoeba-infecting Marseilleviridae (18), the Pithovirus genome sequence either corresponded to a terminally redundant circularly permuted linear DNA molecule or to a closed circle ([Fig. S2](#)), in contrast with the linear Pandoravirus genomes flanked by terminal repeats (9). Finally the Pithovirus genome is predicted to encode a mere 467 proteins, much less than the 2,500 predicted proteins of *Pandoravirus salinus*. The Pithovirus particle appears to

be out of proportion with its gene content compared with other DNA viruses such as the *Phaeocystis globosa* virus packing a similar number of genes into an icosahedral capsid 150 nm in diameter (150-fold less in volume) (19).

Genome Annotation. The genome sequence was analyzed using BLAST (National Center for Biotechnology Information, NCBI) and a combination of motif search and protein-fold recognition methods ([SI Materials and Methods](#)). As is customary on the discovery of the first member of a previously unknown virus family, the proportion of Pithovirus-predicted proteins with recognizable homologs in the NCBI database was low (152/467 = 32.5%). For comparison, it was 60.6% for Mimivirus (2), 41% for Marseillevirus (18), and only 15.7% for Pandoravirus (9). The best matches are distributed almost equally among DNA viruses, bacteria, and eukaryotes, suggesting the absence of close relatives among previously sequenced organisms (Fig. 2*A*). A very similar distribution ($\chi^2 = 0.683$, $P > 0.95$) was computed for the 159 predicted protein validated by their detection in the particle proteome ([Table S1](#)), confirming that the small fraction of database matches was not due to bioinformatic overpredictions (Fig. 2*C*).

Although only one-third of the best database matches corresponded to viral proteins (representing 11% of the total gene content), the absence of genes coding for translation components, ATP-generating enzymes, or related to cell-division confirmed the viral nature of Pithovirus (20). The low level of sequence similarity of these best matches (44% identical residues in average across the highest BLAST scoring segment pairs) ([Tables S1](#) and [S2](#)) as well as their dispersion among different DNA virus families argues against Pithovirus being a member of any one of them. On the basis of this distribution (Fig. 2*B*), the Pithovirus appears globally more similar to Marseilleviridae (19 best hits), then Megaviridae (15 best hits), and then Iridoviridae (10 best hits), all of which are well-established families of icosahedral large DNA viruses. Remarkably, there were only five Pithovirus proteins with their closest homologs in Pandoraviruses. None of these proteins are clearly associated with a functional attribute, except for a remote phosphoglycerate mutase homolog [Pithovirus protein #15 (pv15), 27% identity].

An unusually large fraction of the Pithovirus genome (129 kb, 21.2%) corresponds to multiple regularly interspersed copies of a noncoding repeat, the intricate structure of which produces a unique fractal-like dot-plot pattern (Fig. 3*A* and *B*). These repeats exhibit GC content (23%) much lower than the coding regions (41%), similar to the one of intergenic regions (24%). These repeats are composed of gene-free 2 kb-long tandem arrays of a well-conserved 150-bp palindromic motif (Fig. 3*C*). This motif is not similar to a previously described mobile element and unrelated to the repeats found in some Iridovirus genomes (21) or *Emiliana huxleyi* virus (22). The high-repeat content of the Pithovirus genome decreases its coding density to an unusually low 68%. This value is restored to 85.7% (i.e., 1,048 bp per gene), typical of viruses and prokaryotes, when the repeat moiety is not taken into account. The unique Pithovirus genome structure may result from the multiplication of an invasive selfish DNA sequence or from physical constraints reflecting a specific genome organization, mode of replication, packaging, or transcription. The high copy number of these palindromic repeats contrasts with their known instability in cellular genomes where they tend to be rapidly eliminated after several rounds of replication (23, 24).

Among the 152 predicted proteins with a database match, only 125 (26.7% of the 467 predicted proteins) are associated with functional attributes. Most of them are poorly informative such as protein–protein interaction motifs (e.g., zinc-finger, ankyrin domain, leucine-rich, or collagen triple helix repeats), or motifs involved in various signaling/regulatory pathways (e.g., kinase, phosphatase, GTP-binding) ([Table S1](#)). Typical of large DNA viruses, the dominant functional categories were DNA transcription (17 genes), DNA repair (11 genes, including an ATP-dependent DNA ligase), nucleotide synthesis (7 enzymes, including a ThyX alternative thymidylate synthase), and DNA replication (5 genes).

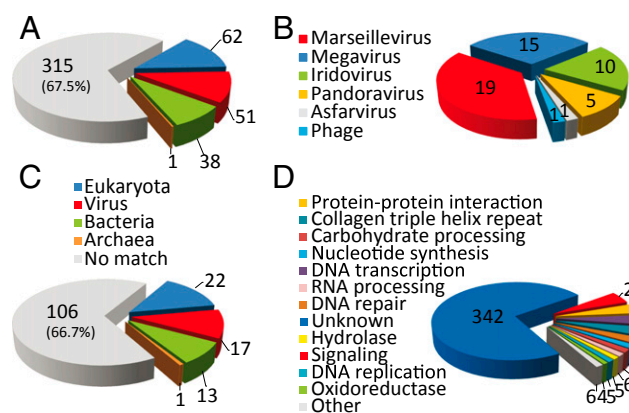


Fig. 2. Distributions of the Pithovirus protein closest homologs. (A) All predicted protein sequences against the NCBI NR (non-redundant) database. (B) Distribution of the 51 best-matching viral proteins. (C) Subset of the 159 proteins detected in the particle proteome. (D) Distribution of the predicted protein functions.

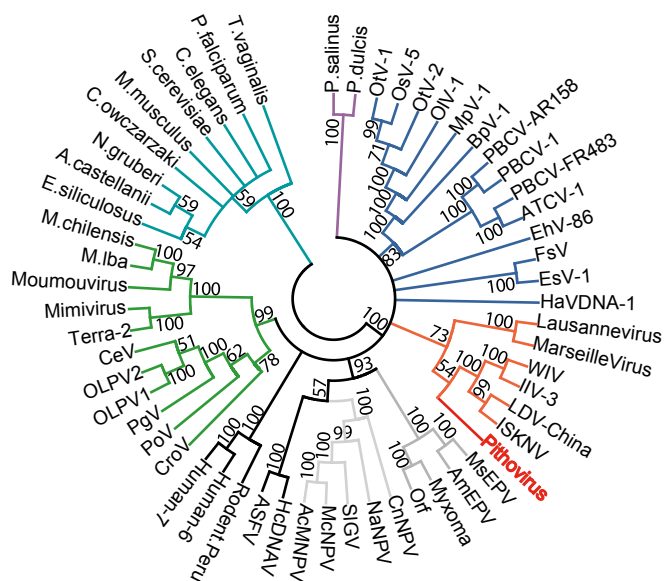


Fig. 4. Clustering of viral and eukaryotic DNA polymerases. A multiple alignment of 57 eukaryotic and large virus DNA polymerase sequences (569 ungapped positions) was computed using the default options of the MAFFT server (40). The neighbor-joining tree was built using the JTT substitution model (estimated $\alpha = 1.05$) and 100 bootstrap resamplings were performed. The tree was rooted at the basis of the eukaryotes and collapsed for bootstrap values <50 before drawing using MEGA5 (41). The Pithovirus DNA polymerase sequence (red) does not cluster with the Pandoraviruses (purple), but falls within a clade clustering the Iridoviruses and Marseilleviruses (orange). Other colors are used to distinguish eukaryotes (turquoise) and viruses from different families: Megaviridae (green), Phycodnaviridae (blue), Herpesviridae (dark gray), Baculoviridae (light gray), Asfar (black), and Poxviridae (gray).

and pv289), oxidoreductases (pv342 and pv467), dehydratase (pv385), and a glycosyltransferase (pv406). These enzymes, as well as 37 *A. castellanii* proteins, all detected in low abundance (Table S1), might be simple by-standers or play a role at the earliest stages of infection.

Transcript Mapping. A whole transcriptome shotgun sequencing analysis was performed on Pithovirus-infected cells at various infection times to validate our gene predictions and assess the transcriptional status of the repeated regions. As expected, a vast majority of the transcribed nucleotides fall in the predicted genic regions. Evidence of transcription was provided for 82% of the predicted coding regions (Fig. 3D), irrespectively of the ORF length (Fig. S5). A residual transcription activity was observed in the intergenic regions and at an even lower level in the repeat-containing regions (Fig. 3D). The sequence tags generated at both extremities of the cDNAs during library construction were used to investigate the 5' and 3' UTR transcripts structures. The 5' extremities were unambiguously mapped for 216 transcripts, leading to an estimated short median length of 8 nt for the 5' UTRs (Fig. S6). Similarly, 3' extremities were reliably mapped for 130 transcripts, estimating the median length of 3' UTRs at 16 nt (Fig. S6). These values are markedly shorter than the ones observed for eukaryotic transcripts, leaving little room for regulatory motifs. The precise mapping of the 3'-transcript ends also demonstrated that 70% are overlapped by a nonconserved palindromic nucleotide sequence capable of forming a hairpin structure that may act as a termination/polyadenylation signal as documented for Mimivirus (31) (Fig. S6).

Discussion

Our discovery of the giant Pandoraviruses, exhibiting astonishing gene contents packaged into uniquely amphora-like shaped particles, hinted that an entire world of viruses remained to be

unrevealed. The simultaneous isolation of two pandoravirus family members (*P. salinus* and *Pandoravirus dulcis*) from two remote locations suggested that they were not rare. The lack of gene content similarity between these Pandoraviruses and the newly isolated Pithovirus is now adding another dimension in the exploration of the world of giant viruses. We now have evidence that viruses from entirely different families (from a genomic point of view) may be associated with the same particle morphology defined by an amphora-like shape, an apex aperture, and a thick electron-dense tegument lined by a lipid membrane enclosing an empty-looking internal compartment. At this point, one may wonder about the diversity of viruses using Pandora box-like particles that remains to be discovered. If this diversity is as large as that of viruses associated with icosahedral capsids, Pandoravirus-like particles might be used to propagate genomes with vastly different sizes and/or molecular structures (e.g., dsDNA, ssDNA, or RNA). The hunt for these viruses is only beginning and speculations about their evolutionary origin should be postponed before many more are found. Unfortunately, there are only three representatives known today (*Pithovirus sibericum*, *P. salinus*, and *P. dulcis*) and they do not share a distinctive set of genes from which to design molecular probes or mine metagenomic data. In the absence of such trademark genes, the identification of additional Pandora box-associated viruses will remain a challenging and serendipitous process.

Beyond their global similarity, a closer inspection of the ultra-structures of the Pithovirus and Pandoravirus particles reveals a number of specific features by which they can be discriminated. These include the spectacular honeycomb-like array of the Pithovirus cork, its striated tegument, and the fuzzy appearance of its surface. Inside their common host, the two viruses also differ by the subcellular structures they generate while replicating. These multiple differences lead us to propose that the uncharacterized KC5/2 endoparasite described more than 10 y ago (32) is a modern relative of the *P. sibericum* rather than another Pandoravirus (9). KC5/2 exhibits a Pithovirus-like striated envelope surrounding a low-density compartment and a protruding cork ("ostiole" in ref. 32) showing parallel stripes. A roll of membrane underneath the cork is also distinguishable (32) (Fig. 14). Finally, cells infected by KC5/2 display Pithovirus-like virion factories ("sol cytoplasm" in ref. 32) surrounded by Pithovirus-like particle components such as stripes, pieces of cork, and amorphous dense material.

Conclusion

Using *Acanthamoeba* as bait, we isolated a large DNA virus from a 30,000 y-old permafrost layer. To our knowledge this is the most ancient eukaryote-infecting DNA virus revived to date. Previous studies reported the detection of genomic signatures of *Emiliania huxleyi* virus in 7,000-y-old sediments underlying the Black Sea (33) and of tomato mosaic tobamovirus in 140,000-y-old glacial ice in Greenland (34), but did not address their viability. Although Pithovirus is the prototype of a new family, its genome structure and replication cycle are similar to that of other large eukaryotic DNA viruses, many of which are human or animal pathogens. Our results thus further substantiate the possibility that infectious viral pathogens might be released from ancient permafrost layers exposed by thawing, mining, or drilling. Climate change in the Russian Arctic is more evident than in many other regions of the world. Whereas the average global temperature has increased by 0.7 °C during the last 100 y, the average temperatures of the surface layer of Arctic permafrost have increased by 3 °C during the same period. In the 20th century, the permafrost in the Northern Hemisphere has diminished by 7% (35). This no doubt corresponded to a large release of micro-organisms from previously frozen soils, an unknown fraction of which was revived upon thawing. Indeed, pathogenic bacteria can survive under low temperatures recurrently causing diseases in circumpolar regions (35). The rich mineral resources and oil reserves of the arctic regions are under increasing pressure for their industrial exploitation (involving mining and drilling). It thus becomes urgent to examine which viruses are expected to

be encountered not only near the surface, but in increasingly deeper and more ancient permafrost layers (down to 3 My) (36, 37). Attempting to isolate and revive increasingly ancient amoeba-infecting viruses from permafrost is an inexpensive and safe way to realistically assess the threat posed by pathogenic animal DNA viruses eventually released from soils and sediments frozen in a distant past. Metagenomic analyses will also be used to safely document the eventual presence of known viral pathogens in permafrost samples containing viable amoeba-infecting viruses.

Materials and Methods

Sample Recovery and Radiocarbon Dating. The P-1084 T permafrost sample was collected in 2000 from the frozen outcrop walls in Chukotka, on the Stanchikovskiy Yar, a steep bank 23 m above the level of the Anui River. It had never thawed. See *SI Materials and Methods* for details.

P. sibericum Isolation and Production. The buried soil sample was resuspended in Prescott and James medium (38) and used to infect *A. castellanii* (Douglas) Neff (American Type Culture Collection 30010TM) cells. The wells presenting an infected phenotype were recovered and used to infect four T-75 tissue-culture flasks plated with fresh *Acanthamoeba* cells. After lysis completion, the virus was purified on a discontinuous sucrose gradient.

Genome Sequencing, Assembly, and Annotation. Around 10 μ g of genomic DNA was recovered from purified particles by combining a phenol/chloroform extraction and purelink genomic DNA extraction mini kit (Life Technologies) protocols. An Illumina 100-bp paired-end library was prepared for sequencing. Another Nextera mate pair library was also prepared and sequenced using 250 bp-length read chemistry on a paired-end flow cell on

the Illumina MiSeq (Illumina). *SI Materials and Methods* contains further details on the bioinformatic genome assembly and annotation procedures.

Transcriptome Study. *A. castellanii*-adherent cells (1.75×10^8) were infected by Pithovirus with 30 particles per cell. For each infection time (4, 11, and 19 h pi) three-fourths of the cells were kept aside for RNA extraction, and the rest were pooled to make a fourth point corresponding to the mix of T4-11-19 h pi. RNA was extracted and quantified and full-length cDNA synthesis was performed using the switching mechanism at 5' end of RNA transcript cDNA library construction kit. The Advantage 2 PCR Kit (Clontech) was used to produce the dsDNA. Illumina 250-bp paired-end libraries were prepared for sequencing. *SI Materials and Methods* contains experimental details and the bioinformatic procedures used for data analyses.

Proteome Analysis. Purified particles (10^8) were resuspended in 100 μ L lysis buffer Tris/20 mM CaCl_2 /1 mM (pH 7.4) and incubated 5 min at 95 $^\circ\text{C}$. Proteins were stacked in the top of a 4–12% (wt/vol) NuPAGE gel (Invitrogen) before R-250 Coomassie blue staining. The gel band was manually excised and cut in pieces before nano-liquid chromatography-tandem MS analyses. The data were processed automatically using Mascot Daemon software (Version 2.3.2; Matrix Science) and the results filtered using the IRMA software (39) (Version 1.31.1). Detailed procedures are presented in *SI Materials and Methods*.

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